Project Brief: Business Problem, Success Criteria and Question Analysis

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1. Business problem

The problem. A hospital records every encounter, its cost, who paid and which procedures were done, but its managers cannot easily see where activity is changing, where cost is concentrated, and how much of it payers actually cover. This project builds a dashboard on the hospital's encounter data to answer those questions.

Why it matters. Payers cover only 30.63% of total claim cost, and 48.71% of encounters show no payer coverage at all. Gaps like these decide how much cost the hospital must recover from patients or absorb, and they affect staffing, billing and planning.

Intended user. Hospital operations and finance managers, for example the people responsible for patient access, billing and capacity planning.

Decisions the findings can support.

- Where to audit claims, starting with insured encounters that recorded no coverage.
- Whether to expand eligibility screening and financial assistance for uninsured patients.
- Which encounter classes and procedures to review for cost.
- How to plan capacity, using typical stay length rather than a distorted average.

Limitations of the evidence. The data covers one hospital and an adult-only population. 2022 holds only five weeks. Zero coverage for an insured patient, zero claim costs and a few very long stays have no explanation in the data. Only 52.6% of encounters have procedure records. The findings describe what was recorded and make no clinical claims.

2. Success criteria

#	Criterion	Measure	Outcome
1	Headline calculations reconcile with an independent check	5 of 5 figures recalculated in Excel match Power BI (27,891 encounters; 101,514,375.52 total claim cost; 13,586 zero-coverage encounters; 48.71%; 75 long stays)	Met
2	Totals respond correctly to filters	7 of 7 slicer tests pass, including across pages	Met
3	Every KPI is documented well enough for another analyst to reproduce	Each KPI has a definition, DAX measure, source columns and caveat in the KPI Definition Sheet	Met

#	Criterion	Measure	Outcome
4	Every data-quality issue has a recorded decision	14 findings logged, each with evidence and decision, in the Data Quality Assessment	Met
5	The dashboard answers at least four of the eight questions in the brief	All 8 answered (questions 1 to 8)	Met
6	No patient names or addresses appear in the dashboard, screenshots or repository	Identifying columns removed in Power Query before loading; raw data files excluded from the repository	Met

3. Question analysis

Question	Metric	Calculation method	Business interpretation	Where shown
1. How have encounter volumes changed over time?	Total Encounters by year	Count of encounter rows grouped by year of the start date	Volumes held between 2,228 and 2,519 a year from 2015 to 2020, with surges of 3,885 in 2014 and 3,530 in 2021. 2022 covers five weeks, so its low count is not a decline.	Overview, Encounter Patterns
2. Which encounter classes account for the largest share?	Class Share %	Encounters in a class divided by all encounters	Ambulatory 44.95%, outpatient 22.59%, urgent care 13.14%, emergency 8.33%, wellness 6.92% and inpatient 4.07%. About two thirds of activity is ambulatory or outpatient.	Overview, Encounter Patterns
3. What proportion of encounters lasted more than 24 hours?	Over 24h %	Encounters where stop minus start exceeds 24 hours, divided by all encounters. All timestamps were valid.	75 encounters, 0.27%. 58 are inpatient, which is 5.11% of inpatient encounters. A few run for months or years, so the median (15 minutes) describes a typical encounter better than the mean (7.27 hours).	Encounter Patterns
4. How do	Total Claim	Sum of total	Ambulatory carries the most cost	Cost &

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total claim costs vary across encounter classes?	Cost by encounter class	claim cost grouped by encounter class, read against each class's share of all encounters	(35.7%), then urgent care (23.0%), outpatient (13.9%), emergency (10.6%), inpatient (8.7%) and wellness (8.1%). Urgent care is only 13.1% of encounters, and inpatient is the costliest per encounter (average 7,761).	Coverage
5. How much payer coverage is recorded, and how many encounters have zero?	Total Payer Coverage, Coverage Rate, Zero Coverage Encounters	Sum of coverage; coverage divided by total claim cost; count of encounters with coverage equal to zero	Payers recorded 31.10 million, 30.63% of claim cost. 13,586 encounters (48.71%) have zero coverage: 8,807 uninsured and 4,779 with an insurer that recorded none.	Cost & Coverage
6. Which procedures occur most frequently?	Procedure Count by description	Count of procedure rows grouped by description, top 10	Routine screenings lead: health and social care needs assessment (4,596), hospice care (4,098) and depression screening (3,614).	Procedures
7. Which procedures have the highest average base cost?	Avg Base Cost (min 20)	Mean base cost by description, shown only where a procedure has at least 20 records	Hemodialysis (about 29,300, 27 records), electrical cardioversion (about 25,903, 1,383 records) and portal vein thrombectomy (about 20,228, 57 records). The costliest procedures are rare, and the threshold of 20 is a judgement.	Procedures
8. How do average claim costs vary by payer?	Avg Claim Cost by payer	Mean total claim cost grouped by payer	Medicaid is highest (6,205) and Dual Eligible lowest (1,696). Uninsured encounters average 5,593.	Cost & Coverage

4. Sources and materials used

- Data: the AnalystLab Africa Week 3 hospital dataset (encounters, patients, procedures, payers, organizations and data dictionary).
- Tools: Power BI, Power Query, DAX and Excel.
- Page background images, logo and Power BI theme file were produced with the help of Claude (Anthropic), an AI assistant. The Poppins typeface used in them is open source.
- Claude was also used for guidance and drafting while building the dashboard and documents. All figures were checked against the data and the analysis decisions are my own.